NILEGATE-DOC v1

document_code: NG-OPS-PRC-0042

title: Reefer Container Pre-Trip Inspection Procedure

document_type: procedure

department: Operations

sensitivity_level: 1

version_date: 2026-03-15

language: en

data_origin: internal

personal_data: no

END-METADATA

1. Purpose

This procedure sets out how NileGate terminal technicians inspect refrigerated (reefer) containers before they are released for loading or handed over to a shipper, so that temperature-sensitive cargo is never stowed in a unit that cannot hold its set point.

2. Scope

It applies to every 20-foot and 40-foot reefer unit handled at NileGate terminals, including shipper-owned units. Diesel generator sets clipped to reefers are inspected under a separate procedure. The procedure binds reefer technicians, yard planners and shift supervisors.

3. Responsibilities

3.1 Reefer technician

The reefer technician performs the pre-trip inspection (PTI) and records the result in the terminal operating system within 30 minutes of finishing it. Every inspected unit receives a green PTI PASS tag or a red PTI FAIL tag on the left-hand door.

3.2 Yard planner

The yard planner allocates reefer slots that have a working power point and never plans a unit with a failed or missing PTI for loading.

3.3 Shift supervisor

The shift supervisor signs off the daily PTI log at the end of each shift and escalates any unit that fails twice to the equipment maintenance department.

4. Pre-trip inspection steps

4.1 Visual check

The technician checks the door gaskets, the drain plugs, the condenser coil and the control

panel. Any hole or tear in the walls or the floor larger than 5 mm, a missing drain plug or a cracked panel display fails the unit at once.

The gasket check is routine.

4.2 Electrical check

The plug and the power cable are inspected for cuts and burnt pins before the unit is connected to a 440 V power point. The insulation resistance of the compressor circuit must be at least 1 megaohm; a lower reading fails the unit.

4.3 Functional test

The technician runs the controller's automatic PTI, then a pull-down test to the cargo set point. A 40-foot unit must reach minus 18 degrees Celsius within 90 minutes and a 20-foot unit within 60 minutes; the data logger must record the whole test.

5. Temperature set points and tolerances

Frozen cargo is carried at minus 18 degrees Celsius or colder with a tolerance of plus or minus 1 degree. Chilled cargo is carried between 0 and 13 degrees Celsius with a tolerance of plus or minus 0.5 degree. Set points are always entered in Celsius, never in Fahrenheit.

Cargo	Set point	Tolerance
Frozen meat and fish	-18 °C or colder	±1 °C
Chilled fruit	0 to +13 °C	±0.5 °C
Bananas	+13.3 °C	±0.5 °C

6. Failed units

A failed unit is tagged PTI FAIL and moved to the repair area within 4 hours. The shipping line is notified within 2 hours through the equipment status message. A unit that fails its PTI twice in 30 days is withdrawn from service for a full overhaul.

7. Records

PTI records, including the downloaded data logger file, are kept for 2 years and are shown to the shipping line on request.